Does Twitter feel the war in real time?

VADER sentiment vs. Russia-Ukraine war events at CHTC scale

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Introduction

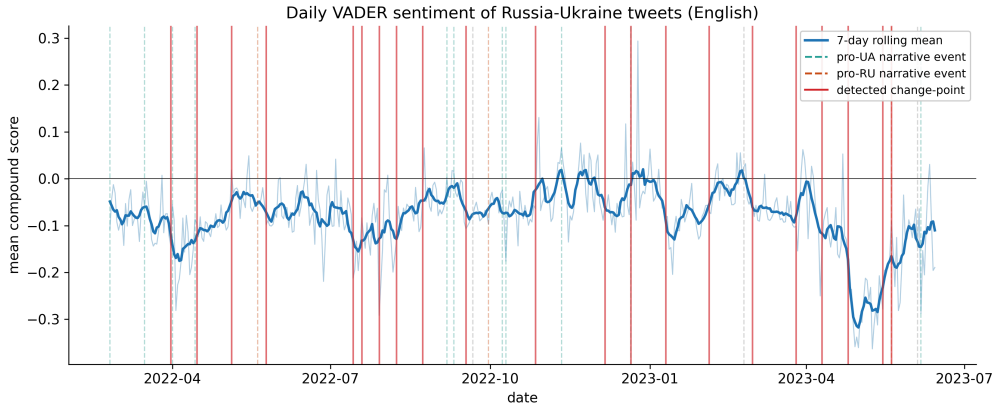
We ask whether public tweet sentiment shifts when a major Russia-Ukraine war event happens. We scored **70.9 million tweets** (~52 GB) from the Kaggle `bwandowando/ukraine-russian-crisis-twitter-dataset` on CHTC with HTCondor, then ran three analyses against 21 curated events: Welch *t*-tests, PELT change-point detection, and OLS regression on event dummies with controls. The model explains ~9% of daily variance ($R^2 = 0.09$, $F = 2.4$, $p = 0.0007$); strongest effects are **Kherson liberated** ($\beta = +0.12$, $p = 0.006$) and **Kakhovka dam** ($\beta = -0.11$, $p = 0.02$). PELT flags 21 breaks; only one matches a labeled event.

Data, computation, results

Data. 480 daily CSVs (~52 GB), Feb 2022 to June 2023. Columns: `text`, `tweetcreatedts`, `language`, `retweetcount`, `followers`. Rows with empty text or bad timestamps dropped (<2%). After scoring: 70,876,101 tweets over 476 days. Primary pool is English (VADER is English-trained).

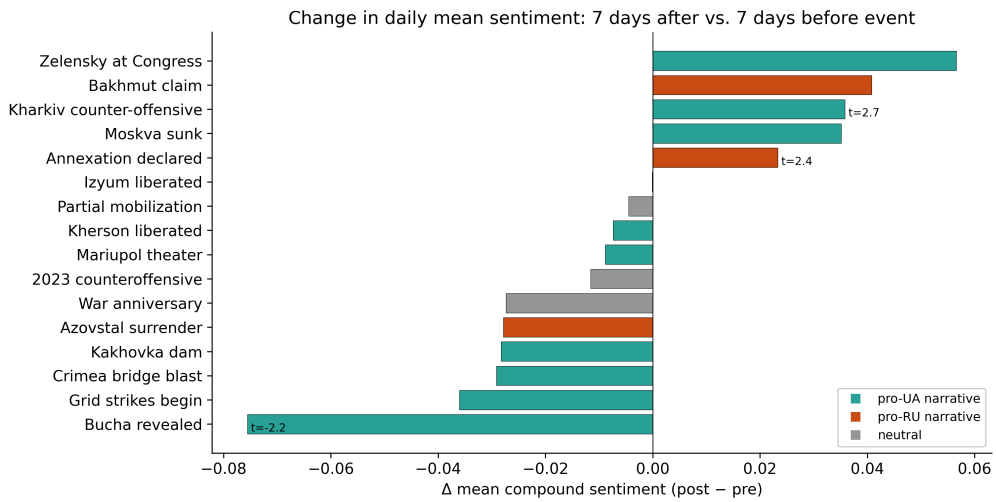
Computation. An Apptainer container (pandas + vaderSentiment, ~340 MB) was staged to `/staging/nomatteson/`. Because `/staging` caps users at 100 inodes, we packed the CSVs into 39 multi-day chunks. For each chunk, `score_tweets.py` streams the file in 200k-row blocks, runs VADER per tweet, and emits one row per (date, language) with sums of compound, compound², pos/neg/neu, followers, and followers×compound — sufficient statistics, so the aggregator never re-reads raw tweets. **Per job:** 1 CPU, ~1.2 GB RAM, 8 GB disk; chunks took 7-50 min. **Total:** 39 parallel jobs, ~1h40min wall-clock. Raw tweets never left `/staging`.

Results.



Daily VADER sentiment with annotated events and detected change-points

English daily mean compound sentiment drifts between -0.05 and -0.15, with shorter negative dips around Bucha (April 2022) and the deepest excursion during the Bakhmut period (spring 2023). Of the 16 events with full [-7, +7] windows, three pass $|t| > 2$: Kharkiv counteroffensive (+0.04, $t = 2.7$), annexation declared (+0.02, $t = 2.4$), and Bucha revealed (-0.08, $t = -2.2$).



Event study: post - pre daily mean, 7-day windows

The OLS regression (`mean_compound ~ t + log(n_tweets) + 17 event dummies`, $n = 476$) gives $R^2 = 0.09$, $F = 2.4$ ($p = 0.0007$). Four coefficients clear $|t| > 2$: Kherson liberated (+0.12, $p = 0.006$), Kakhovka dam (-0.11, $p = 0.02$), anniversary (+0.09, $p = 0.04$), Zelensky at Congress (+0.09, $p = 0.04$). PELT flags 21 breaks; only 2022-12-21 (Zelensky) aligns with a labeled event. Reach-weighting shifts amplitude but not sign.

Weaknesses. (i) VADER is English-only; non-English tweets look falsely neutral. (ii) Twitter users are not the public. (iii) Durbin-Watson = 0.70 means autocorrelation understates OLS standard errors. (iv) Events cluster, so dummies aren't causal. (v) No bot filter.

Conclusion

War events explain ~9% of daily variance on English Twitter, but a handful register: Kherson's liberation and the Kakhovka dam cross $|t| > 2$ with expected sign, and Bucha is the clearest single-event dip. Most change-points don't match labeled events — either

unlabeled drivers or VADER tracking vocabulary drift. Future work: multilingual sentiment (XLM-R), a bot filter, and regression-discontinuity around each event.

Contributions

Member	Proposal	Coding	Presentation	Report
Calvin Sharpe	0	0	0	0
Nils Matteson	1	1	1	1
Pravin Schmidley	0.3	0.1	0.3	0.1
Will Tappa	0.3	0.2	0.3	0.1
Yash Rajani	0	0	0	0

1 = full, 0.1-0.9 = partial, 0 = none.

Clone: `git clone https://github.com/matteso1/STAT405project.git`